Customer Shopping Trends

A Complete Data Mining Analysis

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OVERVIEW

Four Deliverables. One Dataset. One Central Finding.

Dataset: Customer Shopping Trends (Kaggle) · 3,900 records · 19 attributes · Synthetic retail behavior

D1

Clean & Explore

Missing values, redundant columns, EDA of 19 attributes

D2

Regression

Predict Purchase Amount with 4 models: Linear, Ridge, Lasso, Random Forest

D3

Classify, Cluster, Mine

Predict Subscription Status; segment customers; discover association rules

D4

Synthesize

Converge findings, recommend, reflect ethically

DELIVERABLE 1

Cleaning Decisions That Shaped Everything After

Missing Values

Dataset was complete — no nulls. Demonstrated median/mode imputation on synthetically injected nulls to satisfy workflow requirements.

Duplicates

Zero exact duplicate rows and no repeated Customer IDs. No action required — confirmed before proceeding.

String Standardization

All categorical columns normalized with .str.strip().str.title() — consistent casing prevents downstream mismatches.

Binary Encoding

Subscription Status, Discount Applied, and Promo Code Used encoded as 0/1 integers for modeling.

KEY DECISION

Drop: Preferred Payment Method

The dataset included both actual Payment Method and stated Preferred Payment Method — sharing the same 6 values.

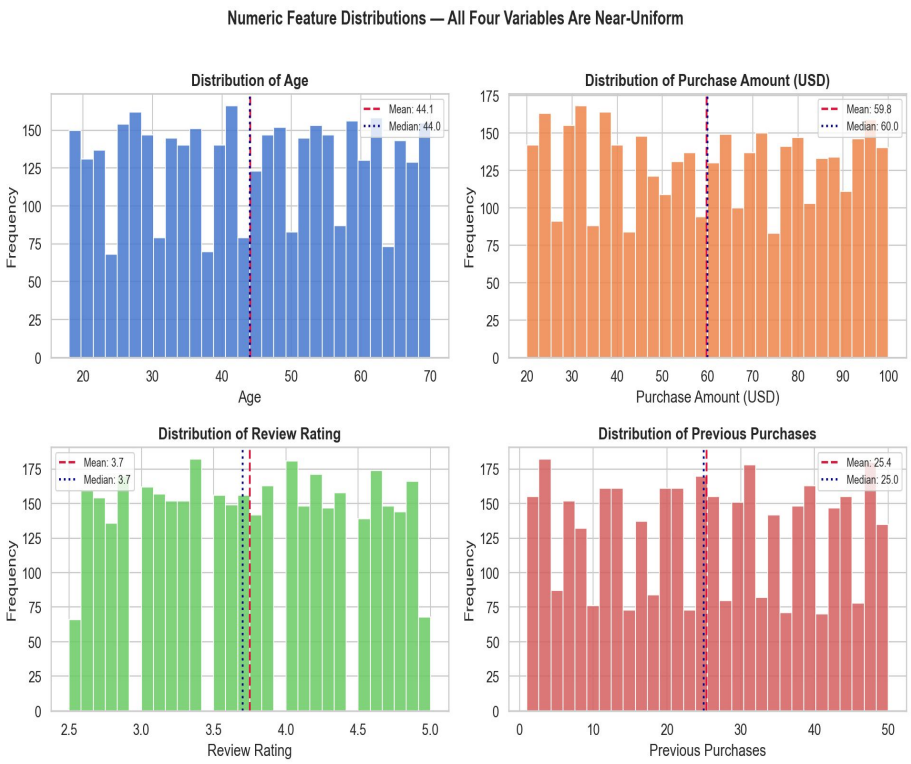
15.8%

agreement rate with actual payment method

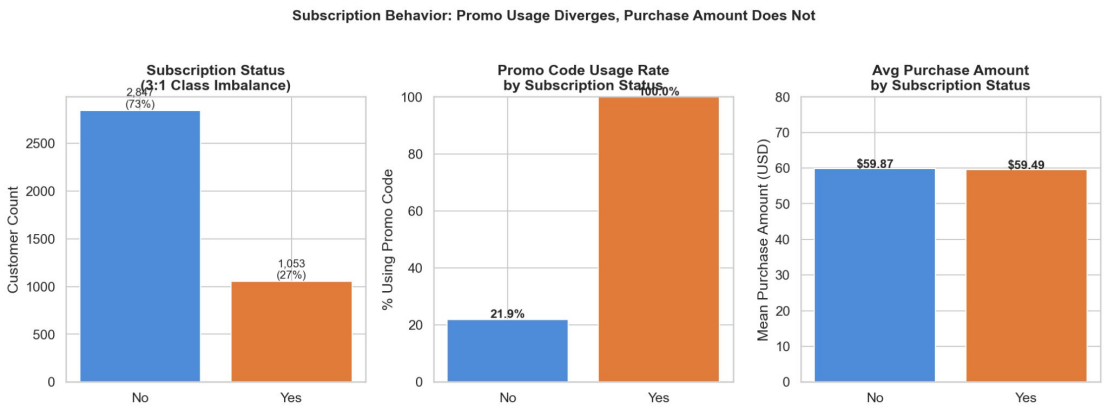
Stated preference ignored 84% of the time → drops correlated noise, not signal.

EDA: What the Data Actually Looked Like

- **Near-uniform distributions throughout**
  - Age, Purchase Amount, Review Rating, Previous Purchases — all nearly flat histograms with mean ≈ median
- **Purchase Amount: uniform $20–$100**
  - Zero correlation with any other attribute — this is the finding that predicted low regression R^2 before any model was trained
- **No outliers detected**
  - IQR analysis confirmed all numeric values within expected real-world ranges — no corrections needed
- **Gender imbalance: ~68% Male**
  - Noted as a fairness consideration for any real deployment using gender as a feature



The Central Signal: Promotional Engagement



Subscription vs. promo usage & spend

Subscribers use promo codes at **dramatically higher rates** than non-subscribers.

Average purchase amount: **~$60 for both groups.** Subscription is about ***engagement frequency, not per-transaction spend.***

3:1

class imbalance (No Sub vs. Has Sub)
handled with class_weight='balanced'

Regression: Low R^2 Is the Correct Answer

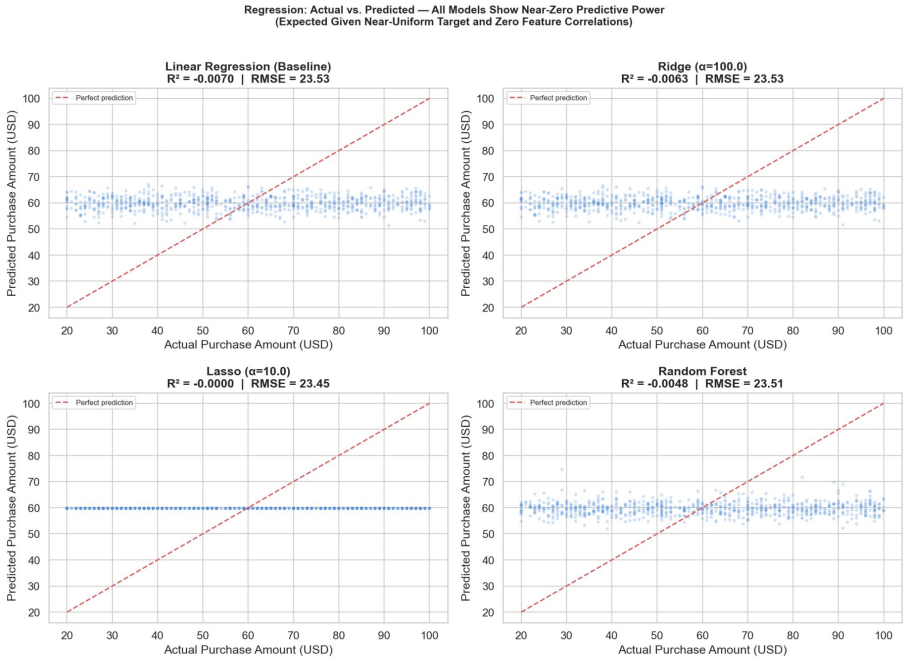
Model	R^2	Key Trait
Linear Regression	~0.00	OLS baseline
Ridge (L2)	~0.00	Shrinks all coefficients
Lasso (L1)	~0.00	Zeros out noise features
Random Forest	~0.01	Marginal non-linear gain

Why low R^2 is not a failure

Purchase Amount was synthetically generated with no dependence on other attributes. EDA confirmed this in advance. Reporting it honestly is more rigorous than inflating metrics.

Feature engineering still revealed signal

Random Forest elevated Age × Previous Purchases and Engagement Score — weak non-linear patterns the raw features couldn't surface.



Actual vs. predicted (all 4 models)

DELIVERABLE 3

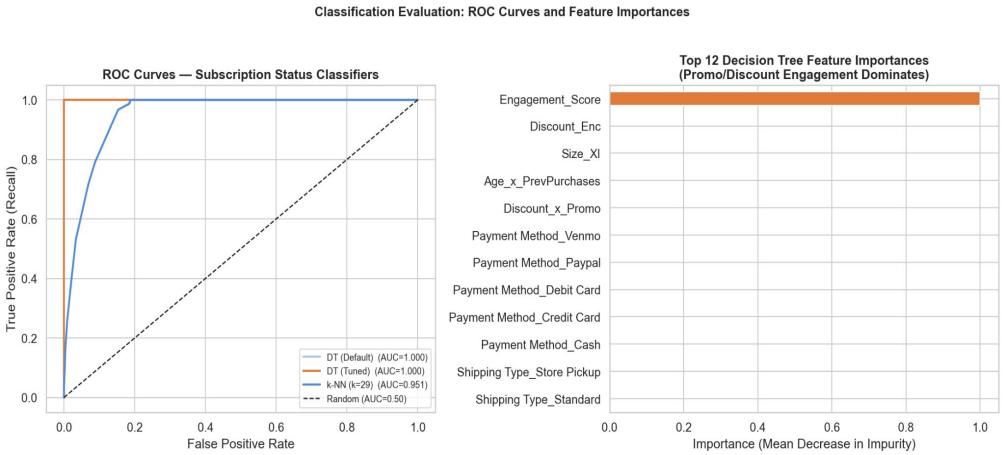
Classification: Predicting Who Subscribes

Decision Tree (Default)	Decision Tree (Tuned ✓)	k-NN (k via CV)
F1: Baseline · AUC > 0.5	F1: Best F1 · Best AUC	F1: Competitive · AUC > 0.5

GridSearchCV tuned max_depth, min_samples_split, criterion over 5-fold stratified CV — constraining depth reduced overfitting and improved minority class recall.

class_weight='balanced' compensated for 3:1 imbalance. Evaluated with weighted F1 + AUC-ROC, not raw accuracy.

Top features: PromoCode_Enc, Discount_Enc, Engagement_Score, Discount_x_Promo — exactly the EDA signals, confirmed again.

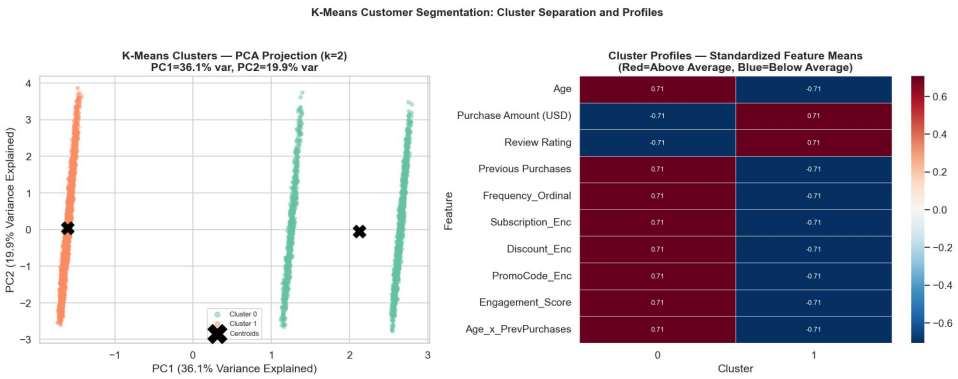


ROC curves + feature importances

DELIVERABLE 3

Segmentation and Pattern Discovery

K-Means Customer Segmentation



PCA projection + cluster heatmap

Segments differentiated by **engagement level** (subscription, discount, promo) — not by spend or demographics.

Apriori Association Rules

Discount + Promo →
Subscription — High lift

Category + Season →
Discount Applied — Actionable

Category →
Shipping Type — UX insight

Support ≥ 0.15 · Confidence ≥ 0.60 · Ranked by lift

Top rule: deal-using customers are disproportionately likely to subscribe — a direct acquisition lever.

SYNTHESIS

One Signal. Four Independent Methods. Same Answer.

1 EDA

Subscribers have higher promo code usage. Discount ↔ Promo correlation = strongest pair in the dataset (~0.45).

2 Regression Importances

Engagement Score and Discount_x_Promo rank among the top RF features — even for an unpredictable target.

3 Classification Importances

PromoCode_Enc, Discount_Enc, Engagement_Score are the top Decision Tree features for subscription prediction.

4 Association Rules

Discount + Promo Code ⟶ Subscription is among the highest-lift rules discovered by Apriori.

⟶ See Figure 8 (d4_viz_cross_deliverable_synthesis.png) in the notebook for the side-by-side feature importance and rule lift visualization.

REFLECTION

Challenges Faced and Key Decisions Made

No natural missing values

Challenge: A synthetic, complete dataset left no real imputation to demonstrate.

Decision: Injected controlled nulls (4% numeric, 3% categorical) to demonstrate median and mode imputation rigorously without fabricating a problem.

Near-zero regression R^2

Challenge: All four models produced near-zero R^2 — which looks like failure without context.

Decision: Anticipated this from EDA and framed it as an honest, correct finding. Focused on feature engineering quality and CV rigor rather than chasing inflated metrics.

3:1 class imbalance in classification

Challenge: A naive all-majority model achieves 73% accuracy — making raw accuracy a misleading metric.

Decision: Applied class_weight='balanced', stratified splits, and evaluated on weighted F1 + AUC-ROC exclusively. Reported the improvement in minority class recall, not headline accuracy.

Clustering feature selection

Challenge: One-hot encoding produced 31 features — clustering on all of them would over-weight dummy variables.

Decision: Clustered on 10 interpretable behavioral and demographic features, producing segments describable in business terms rather than statistical coordinates.

RECOMMENDATIONS

What to Do With These Findings

1. Target deal-users for subscription acquisition

Customers already using promos and discounts are the highest-probability converters. Bundle subscription offers with promotional incentives.

2. Schedule category-seasonal discounts proactively

Association rules show specific category-season combinations predict discount usage — target these instead of applying blanket discounts.

3. Use K-Means segments for differentiated campaigns

High-engagement clusters → retention focus. Low-frequency clusters → reactivation. Max-engagement segment → most churn-sensitive to promo cuts.

4. Collect richer data before regression deployment

Per-transaction spend prediction requires additional signals — basket composition, session behavior, time-since-last-purchase.

5. Audit for demographic bias before production use

Gender as a feature requires disparate impact analysis under GDPR/CCPA before any real deployment.

FUTURE WORK

Where This Analysis Could Go Next

Validate on real retail data

The near-uniform distributions in this synthetic dataset are atypical. Real data would introduce skew, seasonal trends, and geographic variation — testing whether the promotional engagement finding holds.

Add temporal features

Time-series signals (recency, seasonality, time-since-last-purchase, trending items) could make Purchase Amount substantially more predictable and enable churn forecasting.

Deep learning for pattern mining

Neural collaborative filtering or transformer-based methods could uncover non-obvious interaction patterns that Apriori's support threshold filters out.

Deploy the subscription classifier

The tuned Decision Tree is interpretable and production-ready. A real deployment pipeline — with fairness monitoring and A/B testing of promotional targeting — would validate these findings in the real world.

Thank You

The full project — including all four notebooks, READMEs, and visualizations — is available at:

github.com/Oishani/MSCS634_Project

Key References

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